

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			- Biconcave + {increases diffusion/ more oxygen absorbed/ increase gas exchange} / large surface area + {diffusion/ oxygen absorbed/ gas exchange} (1) {Biconcave/ thin} + short diffusion pathway (1) - Enucleate/no nucleus + more space for haemoglobin (1)	3			3		
	(b)			12 or 13 or 14 (µm) = 3 marks If incorrect award 2 marks for answer between 12-15 and shown to a decimal place If both above incorrect award 1 mark for (diameter of B/ diameter of A) × 8 (1)		3		3	3	
	(c)	(i)		100 (2) (45 /4.5 × 10 ⁶) × 10 ⁷ (1)	1	1		2	2	
		(ii)		Yes. Mean volume higher than {normal / 95} (ecf)			1	1		
		(iii)		Any two (× 1) from: A. Fewer cells so {less haemoglobin/ lower total surface area} (1) B. Larger cell therefore {smaller SA : vol / increased diffusion pathway} (1) C. Larger cells so harder for red blood cells to pass through capillaries (1)			2	2		
				Question 4 total	4	4	3	11	5	0